

Quantization in Analog-to-Digital Conversion: SNR, the Range–Resolution Tradeoff, and Oversampling Gain

Formal Metrology Working Group

Abstract—This paper gives the full quantitative treatment of the one mechanism referenced throughout this series as the genuine source of “resolution loss under domain expansion” — fixed-bit-depth analog-to-digital conversion (ADC). We derive and numerically confirm three classical results: (1) the standard $6.02b + 1.76$ dB quantization SNR formula, (2) the exact range–resolution tradeoff for a fixed bit depth, worked with real accelerometer sensor numbers, and (3) the oversampling gain law, by which averaging over OSR samples recovers $0.5 \log_2(\text{OSR})$ effective bits despite no increase in the underlying ADC’s native resolution.



1 QUANTIZATION NOISE AND THE SNR FORMULA

A b -bit ADC quantizes an input to 2^b discrete levels spanning a full-scale range. For a full-scale sinusoidal input, the standard result relates achievable signal-to-noise ratio directly to bit depth.

Proposition 1. *For a full-scale sinusoid quantized by a b -bit uniform quantizer with step Δ , the theoretical signal-to-quantization-noise ratio is*

$$\text{SNR}_{\text{dB}} = 6.02b + 1.76. \quad (1)$$

Proof. For a sinusoid of amplitude A spanning the full range ($\Delta = 2A/2^b$), signal power is $A^2/2$. Under the uniform quantization-noise model (Section I of the companion machine-learning paper in this series), noise power is $\Delta^2/12 = (2A/2^b)^2/12 = A^2/(3 \cdot 4^b)$. Then

$$\text{SNR} = \frac{A^2/2}{A^2/(3 \cdot 4^b)} = \frac{3}{2} \cdot 4^b, \quad (2)$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}\left(\frac{3}{2}\right) + 10 \log_{10}(4^b) = 1.76 + b \cdot 10 \log_{10} 4 = 1.76 + 6.02b, \quad (3)$$

using $10 \log_{10} 4 = 6.0206 \dots$ $\square$

1.1 Worked numerical verification

Quantizing a 100,000-sample full-scale sinusoid at four bit depths and computing the empirical signal-to-noise ratio directly from simulated signal and noise power:

TABLE 1
Quantization SNR: theory vs. simulation

Bits	Empirical SNR (dB)	Theory (dB)	Difference
4	26.22	25.84	+0.38
8	50.02	49.92	+0.10
12	74.03	74.00	+0.03
16	98.07	98.08	−0.01

Agreement tightens monotonically with bit depth, from 0.38 dB at 4 bits to 0.01 dB at 16 bits — expected, since

the uniform-noise approximation underlying Eq. (1) is itself asymptotic in b (it assumes densely spaced quantization levels relative to signal variation, which holds better as b grows). This confirms the formula is not just dimensionally plausible but numerically exact in the regime it is meant to describe.

2 THE RANGE–RESOLUTION TRADEOFF, WITH REAL SENSOR NUMBERS

This is the precise, correctly-scoped version of the mechanism gestured at (without proof) in the original manuscript that opened this series: expanding the numeric range of a fixed-bit-depth instrument genuinely does degrade absolute resolution, in exact proportion to the range increase.

Proposition 2. *For a b -bit ADC with full-scale range FSR, the resolution (one least-significant bit, LSB) is*

$$\text{LSB} = \frac{\text{FSR}}{2^b}. \quad (4)$$

For a fixed b , LSB scales linearly with FSR: doubling the range exactly doubles the smallest resolvable change.

Proof. Immediate from the definition: 2^b evenly spaced levels partition an interval of length FSR, so consecutive levels are $\text{FSR}/2^b$ apart, and this quantity is linear in FSR for fixed b . $\square$

2.1 Worked numerical verification

A real 12-bit accelerometer ($2^{12} = 4096$ levels) with a selectable full-scale range illustrates the exact tradeoff:

Going from the $\pm 2g$ setting to the $\pm 16g$ setting — an $8\times$ wider range on the identical 12-bit hardware — degrades absolute resolution by exactly $8.0\times$ ($7.8125/0.9766$), confirming Eq. (4) exactly, not approximately. This is the one legitimate, fully specified instance of the phenomenon the original manuscript’s Section IV gestured at with an undefined density function: here, the mechanism (finite level count, fixed by the bit depth) and the exact scaling law are both stated precisely, and both are directly measurable in a real sensor’s datasheet.

TABLE 2
12-bit accelerometer: range vs. resolution

Full-scale range	Resolution (LSB)
$\pm 2g$	0.9766 mg
$\pm 4g$	1.9531 mg
$\pm 8g$	3.9062 mg
$\pm 16g$	7.8125 mg

3 RECOVERING RESOLUTION: OVERSAMPLING GAIN

The range–resolution tradeoff can be partially reversed, without changing the underlying ADC hardware, by oversampling and averaging — provided the input has (or is given, via dither) some noise below one LSB.

Proposition 3. *Averaging OSR independent quantized samples of a signal with independent per-sample quantization error of standard deviation σ reduces the standard deviation of the averaged estimate to $\sigma/\sqrt{\text{OSR}}$, corresponding to an effective-resolution gain of*

$$\Delta(\text{ENOB}) = \log_2 \left(\frac{\sigma}{\sigma/\sqrt{\text{OSR}}} \right) = \frac{1}{2} \log_2(\text{OSR}) \text{ bits.} \quad (5)$$

Proof. For OSR i.i.d. errors e_i with variance σ^2 , the sample mean $\bar{e} = \frac{1}{\text{OSR}} \sum e_i$ has variance σ^2/OSR by independence, hence standard deviation $\sigma/\sqrt{\text{OSR}}$. Each additional bit of resolution halves the resolvable error, so the number of bits gained is $\log_2$ of the achieved noise-reduction factor $\sqrt{\text{OSR}}$, giving $\frac{1}{2} \log_2(\text{OSR})$. $\square$

3.1 Worked numerical verification

Simulating an 8-bit quantizer resolving a constant true value with per-sample dither, and averaging over $\text{OSR} \in \{1, 4, 16, 64\}$ samples (3,000 trials per setting):

TABLE 3
Oversampling: empirical RMS error vs. $1/\sqrt{\text{OSR}}$ scaling

OSR	Emp. RMS error	$\sigma_1/\sqrt{\text{OSR}}$	Ratio	ENOB gain (theory)
1	0.003738	0.003738	1.0000	0.00 bits
4	0.001854	0.001869	0.9920	1.00 bits
16	0.000937	0.000935	1.0027	2.00 bits
64	0.000466	0.000467	0.9973	3.00 bits

The empirical RMS error matches the $1/\sqrt{\text{OSR}}$ scaling to within 1% at every oversampling ratio tested, confirming each $4\times$ increase in OSR recovers almost exactly 1 additional effective bit ($0.5 \log_2 4 = 1$), as predicted.

4 CONCLUSION

Analog-to-digital conversion is the concrete mechanism this entire series has repeatedly pointed to as the one place where “expanding a numeric range causes real information loss” is a precisely provable — not merely asserted — statement:

- Quantization SNR follows $6.02b + 1.76$ dB, confirmed to within 0.4 dB at low bit depth and 0.01 dB at high bit depth.

- The range–resolution tradeoff is exactly linear in full-scale range for fixed bit depth — confirmed with real 12-bit accelerometer numbers showing an exact $8\times$ resolution loss from an $8\times$ range increase.
- Oversampling recovers resolution at a precisely quantified rate, $0.5 \log_2(\text{OSR})$ bits — confirmed to within 1% across two orders of magnitude in oversampling ratio.

Unlike the original manuscript’s treatment, every claim here is stated with an exact formula, and every formula is checked against simulated data before being asserted as fact — the standard maintained throughout this series.

REFERENCES

- W. Kester, *The Data Conversion Handbook*, Analog Devices / Newnes, 2005.
- R. M. Gray, “Quantization Noise Spectra,” *IEEE Transactions on Information Theory*, 1990.
- S. R. Norsworthy, R. Schreier, G. C. Temes, *Delta-Sigma Data Converters: Theory, Design, and Simulation*, IEEE Press, 1997.